

NeuroBridge-S4 Application Narrative

From proxy data demonstration to Artemis II adaptation management

Operational Purpose

The practical purpose of NeuroBridge-S4 is to help HRP convert sparse multimodal astronaut data into individualized adaptation management profiles. The report is not just a statistical output. It answers: which systems are under strain, are changes coherent, is recovery progressing, what should be monitored next, and which support category deserves attention during or after the mission?

The project is designed for physical and psychological state management: not diagnosis, but earlier recognition of coordinated stress, sleep, immune, cognitive, and recovery patterns while they are still actionable.

Direct Mapping to Artemis II Study Streams

Artemis II stream	Data meaning	NeuroBridge-S4 role	HRP output
Standard Measures	Blood, saliva, urine, cardiovascular, functional, nutritional, body-system measures.	Reference deviation, baseline imbalance, biomarker domain scores.	Biological state and recovery profile.
ARChER	Actigraphy, sleep, well-being, social interaction, surveys.	Sleep/circadian and behavioral-neurocognitive adaptation layer.	Fatigue, workload, recovery, and psychological monitoring context.
Immune Biomarkers	Stress hormones, immune cell populations, latent virus reactivation.	Stress-immune coupling and immune-shift interpretation.	Immune follow-up and stress-load monitoring priorities.

Individual Adaptation Profile

For each astronaut, NeuroBridge-S4 produces a structured profile: baseline stability, shifted domains, graph-supported coherence pattern, neurocognitive/psychological relevance, recovery slope, monitoring priority, countermeasure consideration, and high-value next measurement. This protects individual trajectories from being lost inside a crew average.

Crew-Level Summary Without False Group Statistics

The crew-level summary identifies shared and divergent patterns without pretending that n=4 supports conventional population claims. HRP receives a map of which systems shifted across individuals, which person shows delayed recovery, which domains act as central graph nodes, and which additional measurements would sharpen interpretation.

Question HRP asks	NeuroBridge-S4 answer
Is this isolated noise or a coherent pattern?	BACI + graph coherence show whether multiple systems are aligned.
Which system should be watched first?	Centrality and active-domain ranking identify sleep, immune, cognitive, metabolic, or recovery priority.
Did the astronaut return to baseline?	Longitudinal delta and recovery slope modules track return-to-baseline.
What should be collected next?	Data-gap layer recommends the highest-value follow-up stream.
How does this relate to psychology?	Only as evidence-based functional interpretation supported by physiology, biomarkers, behavior, and validated self-report.

Management Translation Examples

Coherent pattern	Functional meaning	Monitoring/support priority
Sleep disruption + autonomic strain + cognitive fatigue	Recovery deficit and reduced workload tolerance.	Sleep/circadian support, HRV/autonomic tracking, cognitive monitoring, workload pacing.
Stress biology + immune/inflammatory shift + delayed recovery	Stress-immune coupling.	Immune follow-up, stress-load context review, recovery protocol.
Metabolic shift + fatigue + poor activity recovery	Energy regulation burden.	Nutrition/hydration review, activity pacing, fatigue monitoring.
Sleep + autonomic + self-report strain + cognitive decline	Acute cognitive-perceptual vulnerability context.	Validated psychological check-in, cognitive workload review, sleep/recovery intervention consideration.

How Proxy Data Are Used Responsibly

The proxy data are not presented as Artemis II data. They serve as a testing ground for the workflow: building a real reference cohort, selecting a real n=4 pseudo-crew, computing deviations, generating domain scores, visualizing graphs, testing BACI sensitivity, and producing reports. This prepares the analysis architecture before real mission data become available.

Full Artemis Implementation

Mission phase	Data role	NeuroBridge-S4 analysis
Pre-mission	Personal baseline and variability.	Baseline stability, baseline imbalance, expected range.
Mission/exposure	Stress, sleep, immune, behavioral, operational context.	Delta from baseline, graph-active domains, BACI, monitoring priority.
Immediate return	Acute recovery and re-adaptation.	Early recovery slope and most persistent shifted domains.
Post-mission recovery	Return-to-baseline and delayed recovery.	Domain recovery order, persistent coherence, follow-up recommendations.

Why This Matters

A method that only reports biomarker changes leaves HRP with interpretation burden. NeuroBridge-S4 adds structure: graph context, biological coherence, individualized profiles, and management translation. It does not make unsupported clinical claims; it makes rare human research data easier to interpret, discuss, validate, and act upon.